Multi-Region AI-Powered Banking Application

Engineering Mission-Critical, High-Availability Cloud Infrastructure with Zero-Downtime Tolerance

Architecture: Active-Active DR

Tools: Terraform & Github Actions

Author: Ikenna Anasieze - Cloud & AI Engineer

Repository: Github.com/donaldnaz/Financeapp

1. Overview

Modern banking systems operate under rigorous availability frameworks. Even brief structural disruptions can inflict significant commercial damage, invite regulatory penalties, and compromise consumer confidence. This project documents the end-to-end design, implementation, and orchestration of a resilient, AI-powered banking application deployed on Amazon Web Services (AWS).

The system features a multi-region deployment architecture designed to guarantee immediate failover recovery without human intervention, ensuring continuous availability during widespread cloud outages. Every infrastructure asset is declared as code, allowing rapid deployment, predictable testing environments, and automated decommissioning.

2. Problem to Solve

Traditional financial monolithic systems and naive single-region cloud architectures are vulnerable to systemic regional downtime, or regional cloud service disruptions. The design criteria for this system were framed around eliminating two primary vulnerabilities:

- **Recovery Time Objective (RTO):** The maximum acceptable duration of downtime before service restoration. In mission-critical environments, the target metric must approach RTO → 0.
- **Recovery Point Objective (RPO):** The maximum acceptable age of data that can be lost due to an outage. For financial ledgers, transactional consistency requires an explicit target of RPO = 0.

3. Technology Stack

The stack incorporates specialized tools segmented into logical infrastructure layers to ensure structural rigidity and separation of concerns:

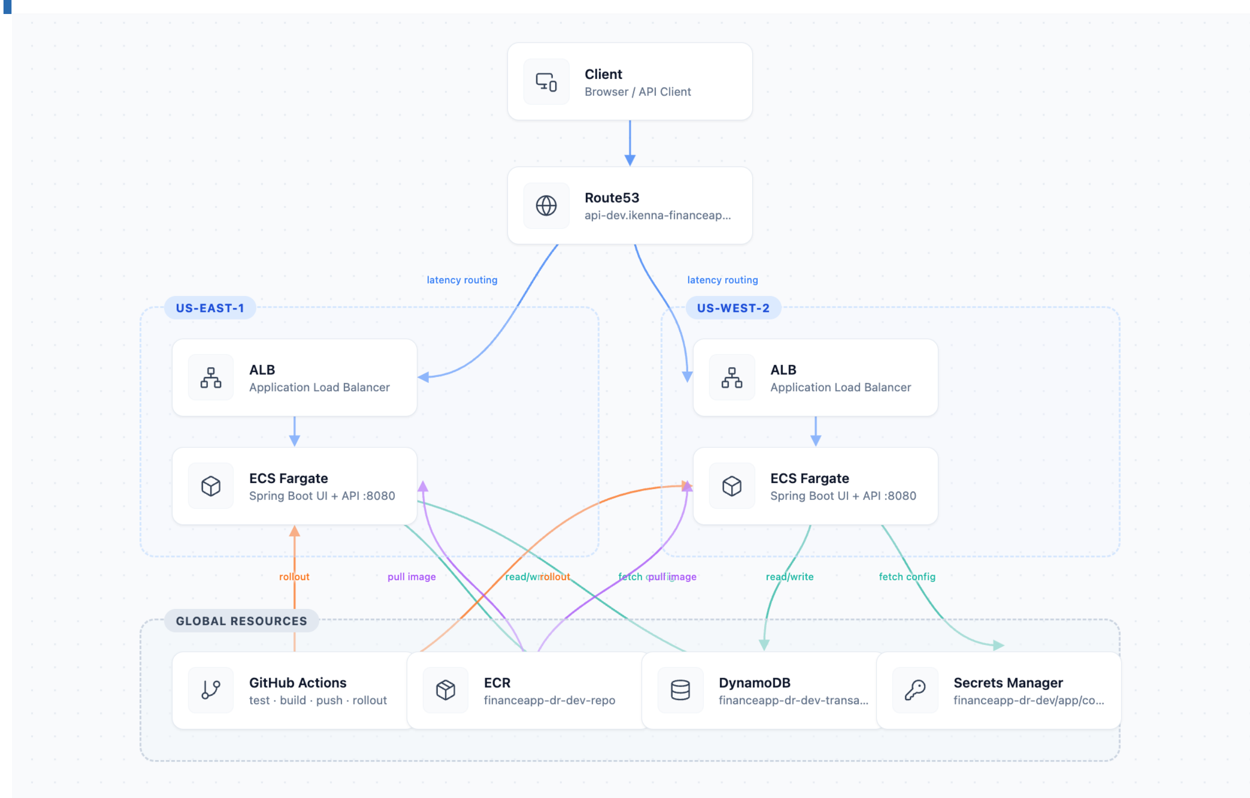
Infrastructure Layer	Technology Utilized	Functional Role
Cloud Provider	*Amazon Web Services (AWS)*	Global cloud infrastructure, cross-region networking, and managed abstractions.
Infrastructure as Code	*Terraform*	Declarative state files, modular network blocks, and multi-provider regional targeting.
CI/CD Engine	*GitHub Actions*	Automated build execution, application containerization, and structural testing

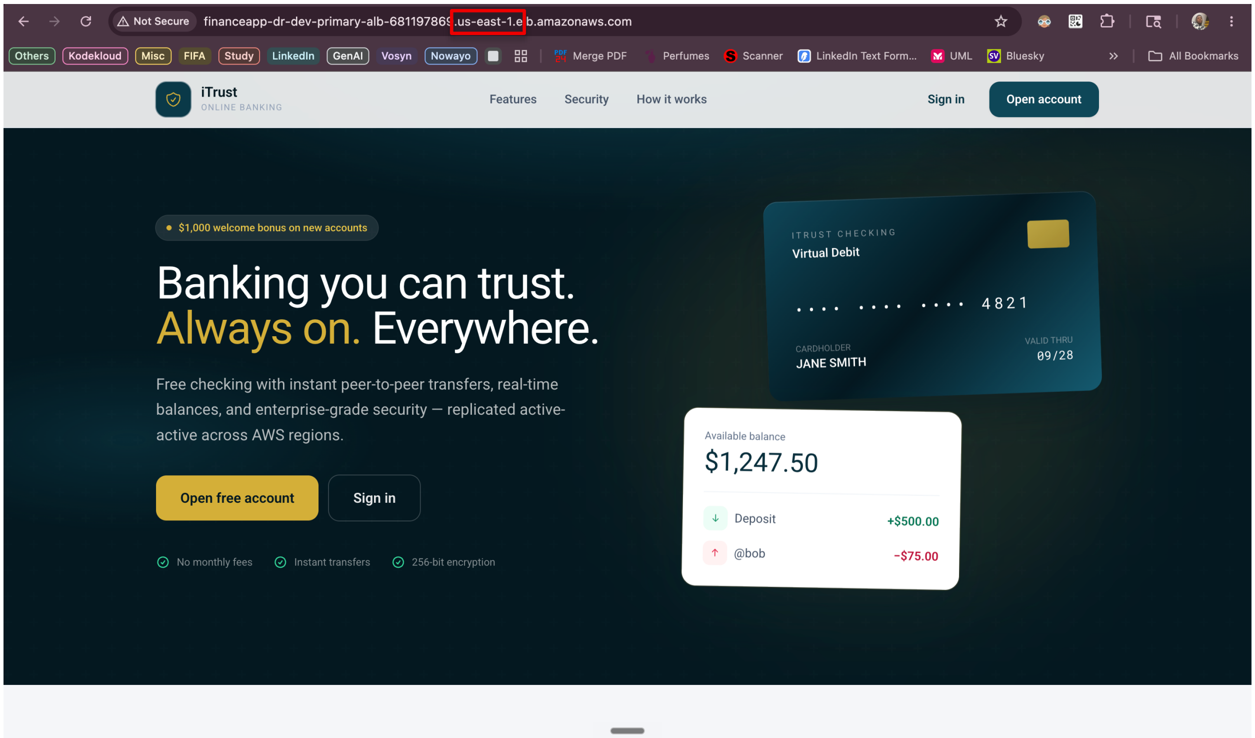
		pipelines.
Compute Tier	*AWS ECS / Fargate*	Serverless container execution, horizontal scaling, and isolated runtime layers.
Data Storage	*Amazon DynamoDB*	Global Tables implementation providing active-active cross-region multi-master synchronization.
Traffic Management	*AWS ALB & Route 53*	routing, continuous health profiling, and DNS failover record automation.
Artificial Intelligence	*Cursor IDE*	Natural Language processing for conversational banking, anomaly detection, and insights.

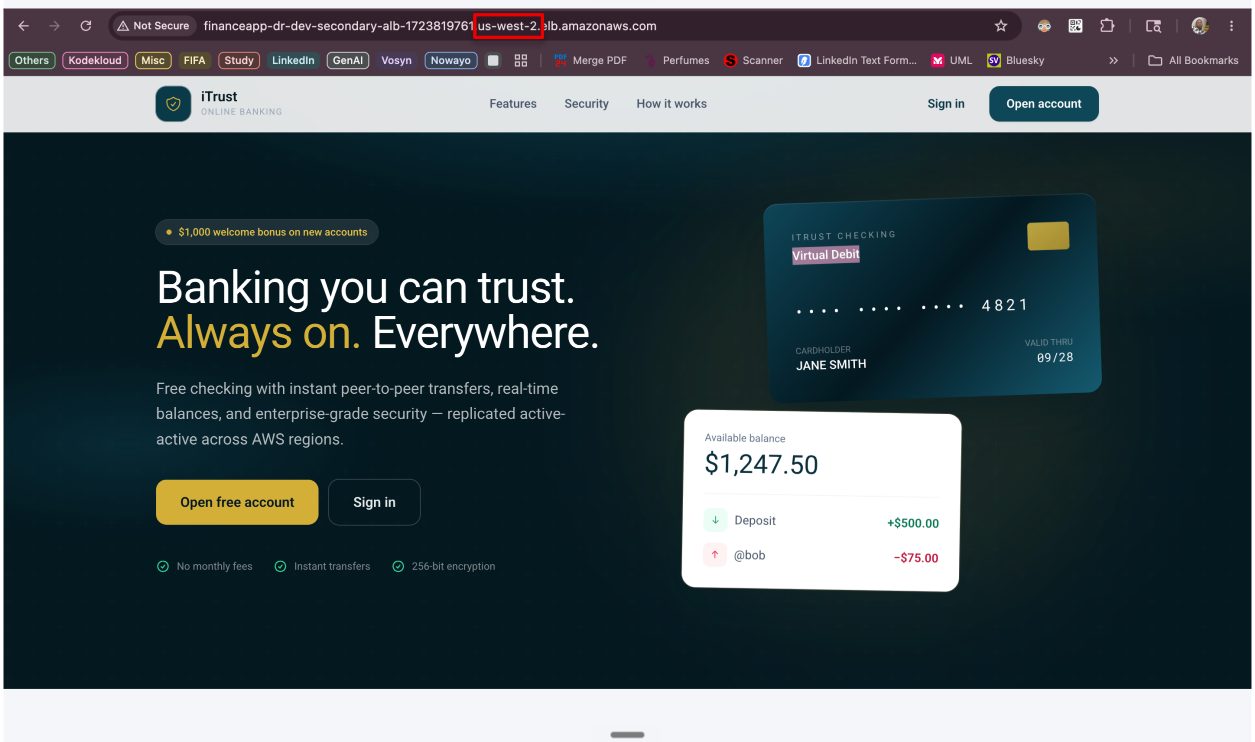
4. Architectural Flow

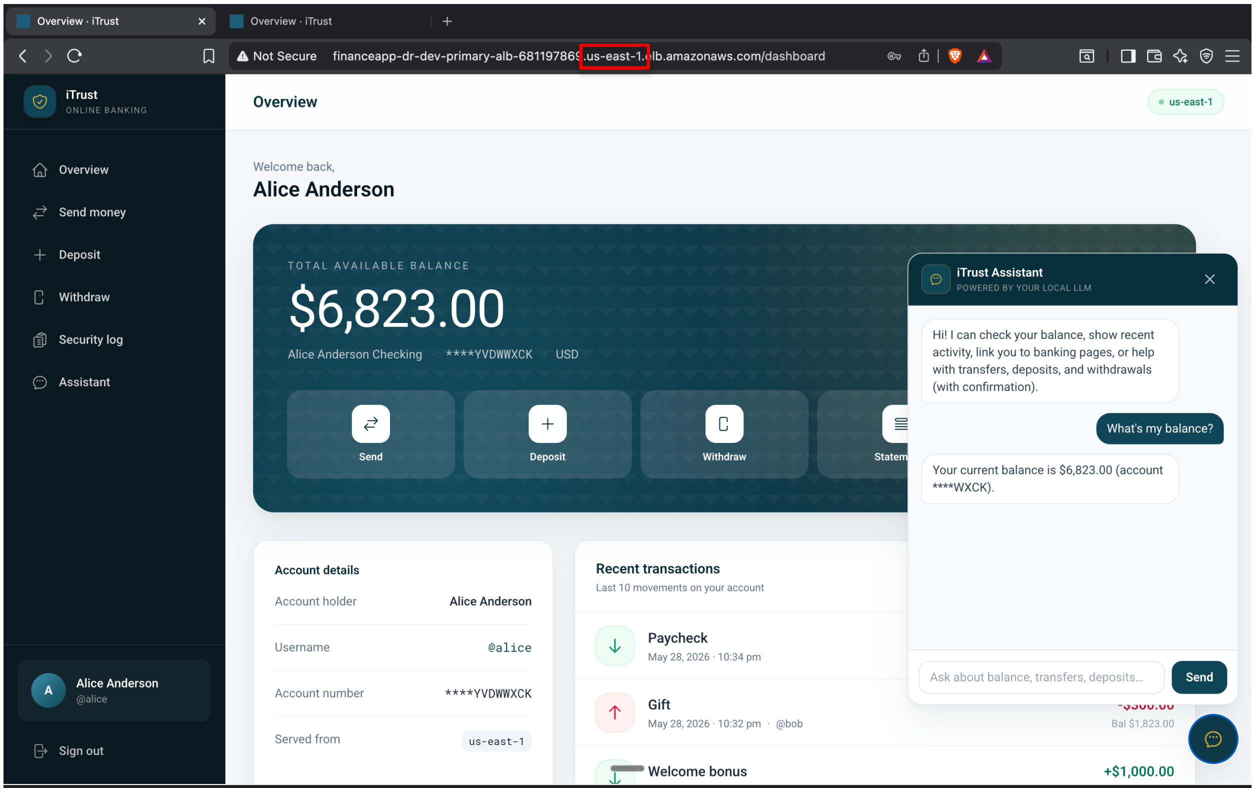
The architecture establishes isolation across two distinct geographic zones: us-east-1 (Primary Region) and us-west-2 (Secondary Region).

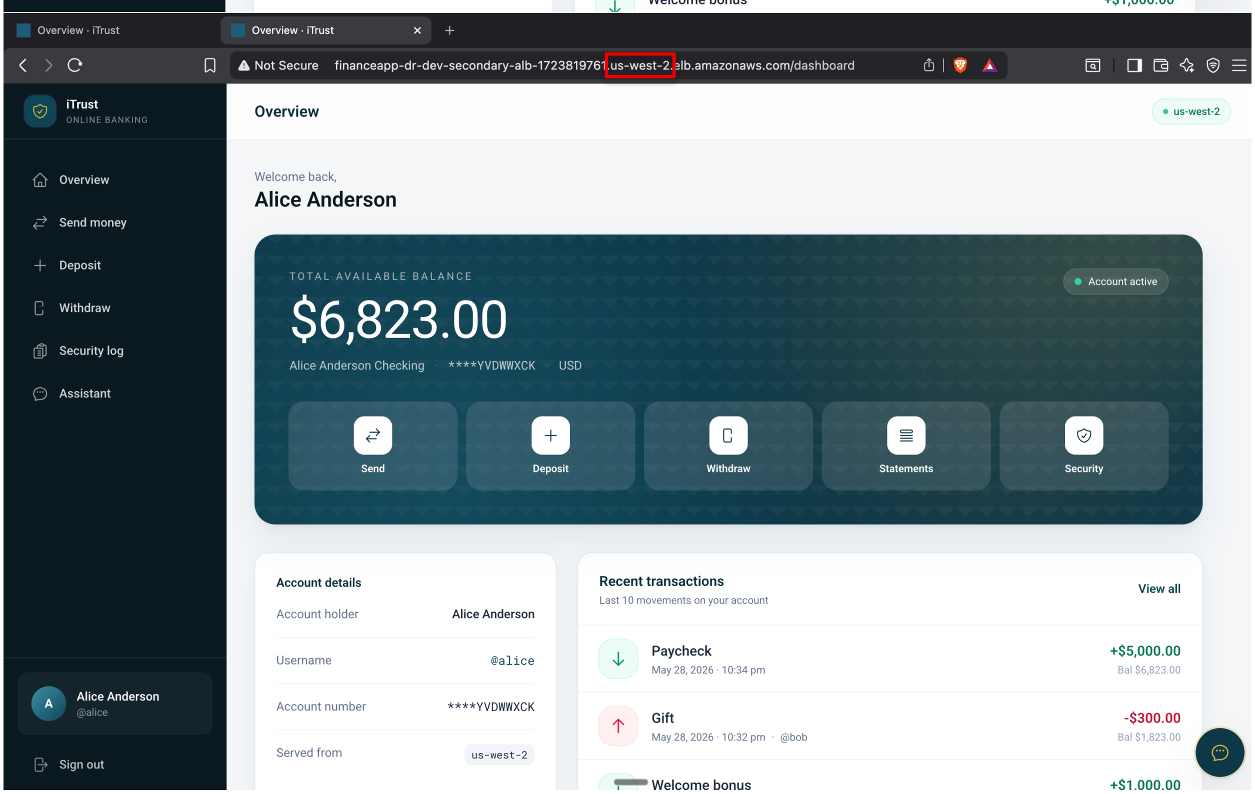
> Client Request → Route 53 (Health-Monitored) → Regional ALB → ECS Fargate Microservices → DynamoDB Global Tables ↔ Multi-Region Cross-Replication.

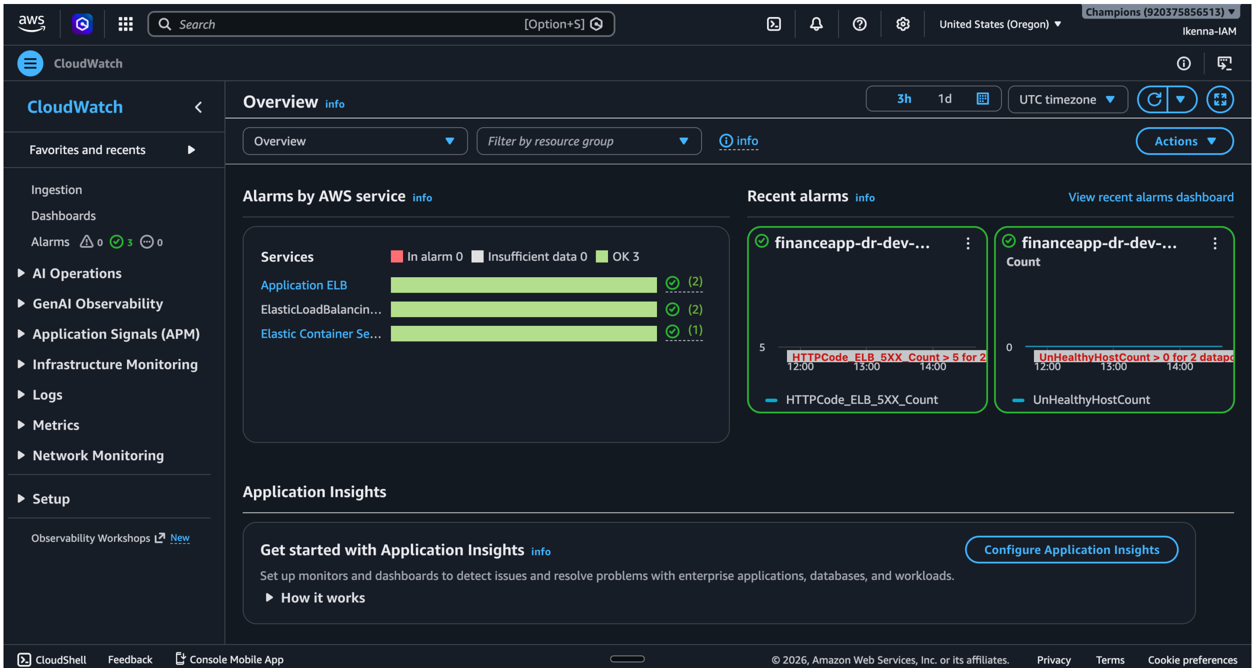



Not Secure
financeapp-dr-dev-primary-alb-681197869.us-east-1.elb.amazonaws.com
Others
Kodekloud
Misc
FIFA
Study
LinkedIn
GenAI
Vosyn
Nowayo
Merge PDF
Perfumes
Scanner
LinkedIn Text Form...
UML
Bluesky
All Bookmarks
iTrust
ONLINE BANKING
Features
Security
How it works
Sign in
Open account
$1,000 welcome bonus on new accounts
Banking you can trust.
Always on. Everywhere.
Free checking with instant peer-to-peer transfers, real-time balances, and enterprise-grade security — replicated active-active across AWS regions.
Open free account
Sign in
No monthly fees
Instant transfers
256-bit encryption
ITRUST CHECKING
Virtual Debit
•••• •••• •••• 4821
CARDHOLDER
JANE SMITH
VALID THRU
09/28
Available balance
$1,247.50
Deposit
+$500.00
@bob
-$75.00

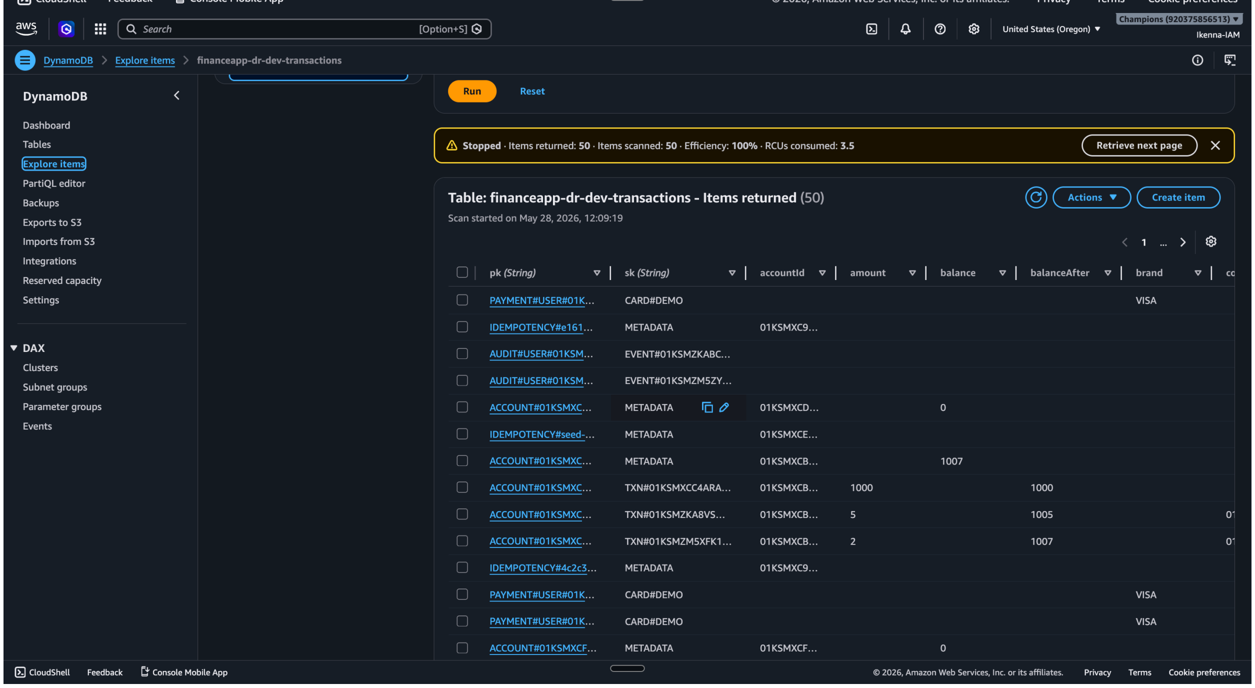

Not Secure
financeapp-dr-dev-secondary-alb-1723819761.us-west-2.elb.amazonaws.com
Others
Kodekloud
Misc
FIFA
Study
LinkedIn
GenAI
Vosyn
Nowayo
Merge PDF
Perfumes
Scanner
LinkedIn Text Form...
UML
Bluesky
All Bookmarks
iTrust
ONLINE BANKING
Features
Security
How it works
Sign in
Open account
$1,000 welcome bonus on new accounts
Banking you can trust.
Always on. Everywhere.
Free checking with instant peer-to-peer transfers, real-time balances, and enterprise-grade security — replicated active-active across AWS regions.
Open free account
Sign in
No monthly fees
Instant transfers
256-bit encryption
ITRUST CHECKING
Virtual Debit
•••• •••• •••• 4821
CARDHOLDER
JANE SMITH
VALID THRU
09/28
Available balance
$1,247.50
Deposit
+$500.00
@bob
-$75.00


Overview · iTrust
Overview · iTrust
Not Secure
financeapp-dr-dev-primary-alb-681197869.us-east-1.elb.amazonaws.com/dashboard
iTrust
ONLINE BANKING
Overview
us-east-1
Overview
Send money
Deposit
Withdraw
Security log
Assistant
Welcome back,
Alice Anderson
TOTAL AVAILABLE BALANCE
$6,823.00
Alice Anderson Checking · ****YVDWWXCK · USD
Send
Deposit
Withdraw
iTrust Assistant
POWERED BY YOUR LOCAL LLM
Hi! I can check your balance, show recent activity, link you to banking pages, or help with transfers, deposits, and withdrawals (with confirmation).
What's my balance?
Your current balance is $6,823.00 (account ****WXCK).
Ask about balance, transfers, deposits...
Send
Account details
Account holder Alice Anderson
Username @alice
Account number ****YVDWWXCK
Served from us-east-1
Recent transactions
Last 10 movements on your account
Paycheck
May 28, 2026 · 10:34 pm
Gift
May 28, 2026 · 10:32 pm · @bob
Bal $1,823.00
Welcome bonus
+$1,000.00
Alice Anderson
@alice
Sign out

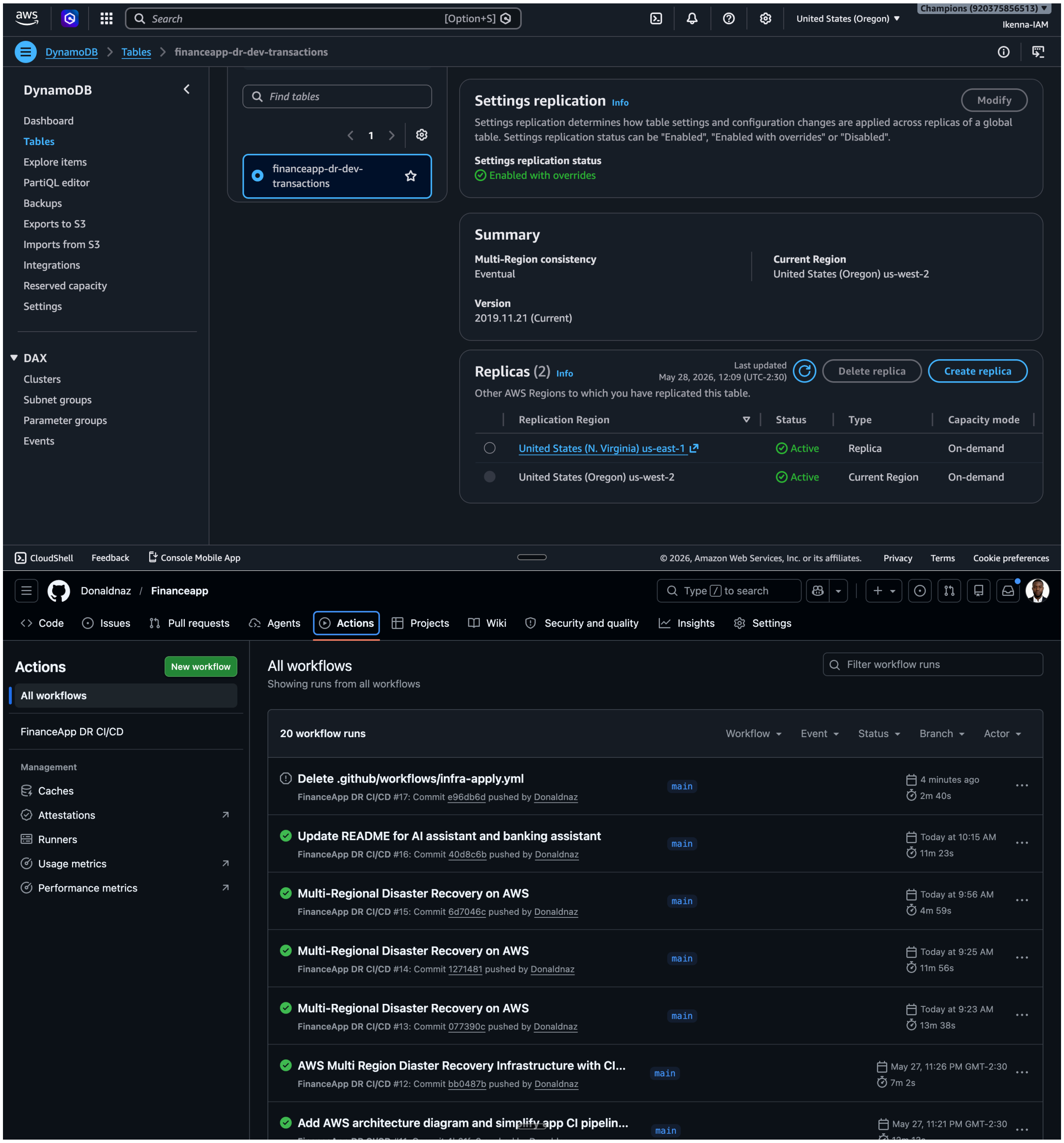

Overview · iTrust
Overview · iTrust
Not Secure
financeapp-dr-dev-secondary-alb-1723819761.us-west-2.elb.amazonaws.com/dashboard
iTrust
ONLINE BANKING
Overview
us-west-2
Overview
Send money
Deposit
Withdraw
Security log
Assistant
Welcome back,
Alice Anderson
TOTAL AVAILABLE BALANCE
Account active
$6,823.00
Alice Anderson Checking · ****YVDWWXCK · USD
Send
Deposit
Withdraw
Statements
Security
Account details
Account holder Alice Anderson
Username @alice
Account number ****YVDWWXCK
Served from us-west-2
Recent transactions
View all
Last 10 movements on your account
Paycheck
May 28, 2026 · 10:34 pm
+$5,000.00
Bal $6,823.00
Gift
May 28, 2026 · 10:32 pm · @bob
-$300.00
Bal $1,823.00
Welcome bonus
+$1,000.00
Alice Anderson
@alice
Sign out



aws
Search
[Option+S]
United States (Oregon)
Champions (920375856513)
Ikenna-IAM
CloudWatch
CloudWatch
Overview info
3h
1d
UTC timezone
Favorites and recents
Overview
Filter by resource group
info
Actions
Ingestion
Dashboards
Alarms 0 3 0
AI Operations
GenAI Observability
Application Signals (APM)
Infrastructure Monitoring
Logs
Metrics
Network Monitoring
Setup
Observability Workshops New
Alarms by AWS service info
Services
In alarm 0
Insufficient data 0
OK 3
Application ELB (2)
ElasticLoadBalancin… (2)
Elastic Container Se… (1)
Recent alarms info
View recent alarms dashboard
financeapp-dr-dev-…
HTTPCode_ELB_5XX_Count > 5 for 2
12:00 13:00 14:00
HTTPCode_ELB_5XX_Count
financeapp-dr-dev-…
Count
UnHealthyHostCount > 0 for 2 datapo
12:00 13:00 14:00
UnHealthyHostCount
Application Insights
Get started with Application Insights info
Configure Application Insights
Set up monitors and dashboards to detect issues and resolve problems with enterprise applications, databases, and workloads.
How it works
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences



aws
Search
[Option+S]
United States (Oregon)
Champions (920375856513)
Ikenna-IAM
DynamoDB > Explore items > financeapp-dr-dev-transactions
DynamoDB
Dashboard
Tables
Explore items
PartiQL editor
Backups
Exports to S3
Imports from S3
Integrations
Reserved capacity
Settings
DAX
Clusters
Subnet groups
Parameter groups
Events
Run
Reset
Stopped · Items returned: 50 · Items scanned: 50 · Efficiency: 100% · RCUs consumed: 3.5
Retrieve next page
Table: financeapp-dr-dev-transactions - Items returned (50)
Scan started on May 28, 2026, 12:09:19
Actions
Create item
pk (String)
sk (String)
accountId
amount
balance
balanceAfter
brand
PAYMENT#USER#01K… CARD#DEMO VISA
IDEMPOTENCY#e161… METADATA 01KSMXC9…
AUDIT#USER#01KSM… EVENT#01KSMZKABC…
AUDIT#USER#01KSM… EVENT#01KSMZM5ZY…
ACCOUNT#01KSMXC… METADATA 01KSMXCD… 0
IDEMPOTENCY#seed-… METADATA 01KSMXCE…
ACCOUNT#01KSMXC… METADATA 01KSMXCB… 1007
ACCOUNT#01KSMXC… TXN#01KSMXCC4ARA… 01KSMXCB… 1000 1000
ACCOUNT#01KSMXC… TXN#01KSMZKA8VS… 01KSMXCB… 5 1005
ACCOUNT#01KSMXC… TXN#01KSMZM5XFK1… 01KSMXCB… 2 1007
IDEMPOTENCY#4c2c3… METADATA 01KSMXC9…
PAYMENT#USER#01K… CARD#DEMO VISA
PAYMENT#USER#01K… CARD#DEMO VISA
ACCOUNT#01KSMXCF… METADATA 01KSMXCF… 0
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences



5. Business Impact

Transitioning from a traditional cloud posture to this multi-region declarative framework yields clear business and operational advantages:

- **Uninterrupted Business Continuity:** The application achieves true regional disaster isolation. Total failure of an entire AWS data-center hub triggers automated recovery procedures without service interruption for users.

- **Optimized Cost Efficiency:** By utilizing AWS Fargate serverless compute units and on-demand DynamoDB pricing models, the operational maintenance costs for running this multi-region infrastructure baseline are minimized to approximately $8.00 / day during idle testing.
- **Elimination of Manual Operations:** Infrastructure code guarantees environment parity. The exact topology can be reproduced instantly, eliminating manual execution variance or configuration drift during disaster scenarios.